Post Processing Cookbook

⚠ Please read Post Processing Quality Reminders before starting any post processing

Launching an AWS Instance

For first-time AWS setup, see: AWS Setup Instructions

- To launch an AWS instance double click on the icon for the launcher script or run it from the command line.
  - Then, open MobaX and connect to localhost port 5902
  - Open a new terminal in the VNC
- To reconnect to an existing instance, just run the launcher again
- To terminate an instance, go to the launcher window and press 'q' to quit and then press '1' to confirm.

⚠ **ALWAYS TERMINATE YOUR INSTANCE AT THE END OF EVERY DAY**

- **Important**: When terminating an instance, all data that is on that computer will be deleted, unless it is in an S3 bucket (anything under `/mnt/<bucket_name>`) or it is in your persistent volume (`~/volume/<volume_name>`)

Common file path structure

Type	Path
Raw files to be post-processed	`/mnt/ppo-experimental-data/<project/fo/dm<#>_<spec/vis>/<date>/<raw>/<run>/`
Corrected output files	`/mnt/ppo-data-services/post-processing/<project>/fo/dm<#>_<spec/vis><dm>/<date>/corrected/`
Output Extract files	`/mnt/ppo-data-services/post-processing/<project>/fo/extracts_<spec/vis>/dm<#>/`

Finding Files Using the Data Library

For more info on the Data Library, see Introduction to the Data Library

- Files are found using the `dl ls` command, which works similar to standard `ls` but will also by default display some extra information about the file's metadata
  - ```
    dl ls /mnt/ppo-experimental-data/SFDCUD_IF/fo/dm21335002_vis/2023-12-09/raw/run_2023-12-09--17-26-46/frame*21414
    ```
  - Use `--filter` or `-w` to filter results on specific metadata values. For example to find all t3st data with material `bone-ca` on a certain day you would use:
    - ```
      dl ls /mnt/ppo-experimental-data/SFDCUD_IF/fo/dm21335002_vis/2023-12-09/raw/run* \
      --filter testplan.mat bone-ca \
      --filter testplan.scan_mode t3st
      ```
    - This is necessary because metadata is no longer stored in the filenames
  - To see only the file path output, use `--bare` or `-b`. This is useful when using the command for running the post-processing tooling or piping into other commands
  - To see all metadata associated with the files, use `--verbose` or `-v`
- Checking # of files
  - ```
    dl ls /mnt/ppo-experimental-data/SFDCUD_IF/fo/dm21335002_vis/2023-12-09/raw/run* \
    --filter testplan.mat bone-ca --bare | wc -l
    ```

Navigating S3 Buckets

- The EC2 instances use `s3fs` to mount the s3 buckets
    ```

- There is a significant slowdown that happens when you `cd` into an S3 bucket on your instance
  - Any commands or scripts you run while cd'ed into an S3 bucket will be much slower
  - e.g. SLOW:

    ```
    cd /mnt/ppo-data-services/post-processing
    ls
    ```

    NOT SLOW:

```
ls /mnt/ppo-data-services/post-processing
```

- Therefore whenever possible, avoid navigating into the S3 buckets, and run commands using full paths. There should really never be a time where cd'ing into a S3 bucket is required
- TIPS:
  - Whenever you would normally use `cd` use `ls` instead
  - Use the `up` and `down` arrows to access previous commands
  - Use `Tab` to autocomplete the navigation, `Tab` twice will the full list of autocomplete options
  - Use `Home` and `End` keys to quickly move the cursor to start or end of the line, this is useful if you want to change your `ls` command to a different one like `pproc`

Task 3V - Vis Correction (ARCHIVED TASK)

Note: This is one of the rare instances where it is necessary to mount the server as writable so use caution.

Single run

- You must be under the date folder and **the script will combine all frames in a run into a single h5 file.** `python ~/ppo/tools/visual/vis_correct_capture.py <file name> --calib_dir <calibration directory>`

Multiple runs

- You must be under the raw folder `python ~/ppo/tools/visual/vis_correct_capture.py <run*> --calib_dir relative --offline`

Additional Details

- `--calib_dir relative` assumes the calibration directory is on the same directory level as the raw directory, otherwise you can supply your own calibration path
- `--calib_dir <path to calibration folder> --bright auto --dark auto --auto_camera_serial` will be used when specifying the exact location of the calibrations to use. When specifying the path, it must include all the folders to reach the final folder with the bright and dark .h5 files.
- `--offline` used when the calibrations are stored in the ppo-server directory structure

Task 3S - Spec Correction (ARCHIVED TASK)

Corrections are only needed for specific data (belt training data and shadow belt data). For all other data, corrections can happen on-the-fly with the various tools

- **Note**: False Positives are corrected on-the-fly during model evaluation.
- Below is an example command for a single run directory or file.
- ```
  python ~/ppo/projects/feasibility_capture/capture_pipeline/correct_capture.py \
  <image dir> \
  --offline \
  --calib_dir relative
  ```
- provided the raw files are in the appropriate dir structure, i.e. `<root dir>/YYYY-MM-DD/raw/`

Task 4V - Vis Annotation and ROI

Step 1. post_processing_aws.py

This script takes raw scan data from our **visual cameras** (VIS) and **hyperspectral sensors** (SPEC), detects specific areas of interest (like green o-rings or ROI squares), and then saves it all for analysis.
  ```

1. Run `python ~/ppo/tools/visual/post_processing_aws.py` or use alias `pproc` on raw **VIS** input files.
2. Use the `dl ls` command within back-ticks to filter the files in the run.

*The most efficient method will be to run the tool **per material**, on a **given dm,** on a **given date** for data collections*

Annotations

```
pproc `dl ls /mnt/ppo-experimental-data/<project ID>/fo/<project code>/<dm#_spec/vis>/<date>/raw/run*/frame*.h5 --filter testplan.mat <material_name> --filter testplan.scan_mode t3st -b`
```

**Path may vary dependent on project*

- This will output annotated vis files, relative spec files, spec and vis pngs, and an info text file under `/mnt/ppo-data-services/post-processing/review/annotation/` folder

**This output can be changed using* `--save_dir` flag

> If for the project, the extract files are not being split into test and validations extracts, and only one extract file is being written per annotation, add the `--no_split` arg.

To find single files, they will be under

```
/mnt/ppo-data-services/post-processing/annotation/<date>/<camera_serial>/<file>
```

ROIs

Same code, but input a list of *train* data instead of *t3st* data

```
pproc `dl ls /mnt/ppo-experimental-data/<project ID>/<dm#_spec/vis>/<date>/raw/run*/frame*.h5 --filter testplan.mat <material_name> --filter testplan.scan_mode train -b`
```

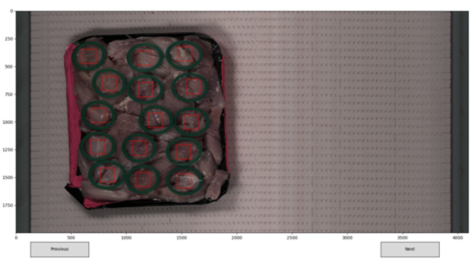
Step 2. review_tool_aws.py

This script is an interactive quality control tool for the data processed by the previous script in Step 1. After the automation identifies features like green o-rings or ROI squares, the review tool lets a DS reviewer visually inspect those results, make adjustments if needed, and confirm the quality before the data is used for model development. *This will make extracts for both spec + vis*

Annotations

```
revt /mnt/ppo-data-services/post-processing/annotation/post_proc_info_<#>.txt --make_extracts
```

review_tool annotation guide



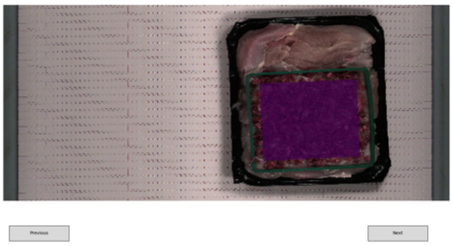
Left click to add or remove annotations (left clicking an existing annotation will remove it). You will be notified when you reach the final page.

Press q to save and exit, or ! to exit without saving. This will overwrite the local annotated files and PNGs with the edited masks (both spec and vis). If a mistake was made after saving, you can re-run the tool, and see the updated masks

ROIs

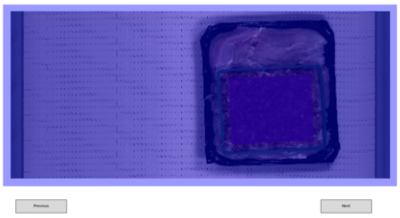
`revt /mnt/ppo-data-services/post-processing/roi/post_proc_info_<#>.txt`

review_tool ROI guide



Navigate between pages using the Next and Previous buttons (or arrow keys)

If the mask displayed in the UI looks incorrect, left click to highlight it blue and mark as redo. Click again to remove the highlight. *Images may already be highlighted in blue, if the automation in step 1 determined it failed*



If the mask displayed in the UI is a bad image and cannot be redone (e.g. bad cropping), right click to highlight red and mark the files as not to be uploaded. Press q to save and exit, or ! to exit without saving.

This will only delete files marked for deleting, it will not edit files marked as redo. Files marked as redo must be done manually using tools/chemometric/polygon_select_ROI.py on the same post_proc.txt file

Re-doing Scans

If any scans need to be redone, rerun `review_tool_aws.py` on the .txt file, and left click to highlight blue the file to be redone

Then run `python ~/ppo/tools/visual/polygon_select_ROI_aws.py` or use alias `vroi` directly on the same post_proc.txt file, and include `--transform` / `--t`

**This will open any files previously marked blue which can then be ROI'd as normal*

- ```
  vroi /mnt/ppo-data-services/post-processing/roi/post_proc_info_<#>.txt --df passthrough --t
  ```

`--t` transforms the vis roi to spec

Task 5V/5S - Vis/Spec TP Extracts, Check Extracts and Training Pixels (ARCHIVED TASK)

If PNGs need to be made manually, use

```
python ~/ppo/tools/chemometric/training_ROI_PNGs.py --file_list /mnt/ppo-experimental-data/<customer>/fo/<DM>/<spec/vis>/corrected/<train scan>*_wMask.h5 \ --vis
```

- Example command (saves `*.png` images in the `vis_training_ROIs` subdirectory)

If extracts need to be made manually use either:

- ```
  python ~/ppo/tools/visual/extract_FO_regions.py --manual <h5 files> --test_split --output_dir <OUTPUT DIR>
  python ~/ppo/tools/chemometric/extract_FO_regions.py --manual <h5 files> --test_split --output_dir <OUTPUT DIR>
  ```
- Extracts created will now be split into validation and testing sets with `_dataset+[val/test]` key value pair in the filename. This means there will be two extract file per annotation file

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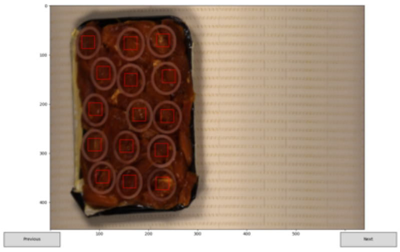
Task 4S - Spec Annotation and ROI

This step is a sanity check to review the spec files. Ideally, **no edits will need to be made during this step**, because the coregistration from vis should be accurate enough that no changes are required

- When reviewing the files, keep in mind the spec image is slightly distorted from the vis image so while some vis extracts will be nice placed in the O-ring, spec may not
- Run `python ~/ppo/tools/visual/review_tool_aws.py` on the exact same info text file using the same command in step 2, but add the `--spec` argument
    - `--spec` - this tag will ensure any changes are only made to the spec file
        - Annotations: `--make_extracts` - similar to vis above, this tag will ensure any changes will result in new extracts being made after the user hits 'q'
        - ROIs: No additional tags are required

review_tool --spec argument



Identical to vis, just displays spec PNGs

- If a mask is edited, it will only overwrite the spec annotated file and PNG, the vis file will not be edited
- Similarly, if an ROI is marked to be skipped for upload, only the spec ROI will be marked, the vis file will remain
    - This is because from the automation algorithm we know that if the vis mask fails, the spec mask will also fail.

Opening and Reviewing PNGs

PNGs will be saved in a subdirectory located where the above command was run. That subdir (for vis) will be called `vis_training_ROIs`.

- You can `cd` to the directory containing the PNGs (ex. `cd mnt/ppo-data-services/post-processing/roi/<date folder>/<camera serial #`), `ls` to view the files, and run `eog <*.png>` to view the output PNGs. Because the image has a mask overlaid, it may difficult to see the FM. If that's the case, you can reference this PNG against the file itself by finding the associated h5 file and opening it in vis_linescan_explorer (alias `vexp/hexp`).

Check PNGs to ensure there is no background material encapsulated in the ROI

- ROIs should **not** contain the following:
    - belt
    - other FOs that are not the FO of interest

NOTE: if ROI'ing densities, the meat background can be included in the ROI as long as the ROId region has the same density of FOs (ie. no huge gaps of *just* meat).

Reviewing Extracts

Extracts can be reviewed using the `~/ppo/tools/visual/vis_linescan_explorer.py` tool, or `vexp/hexp`.

- Extractions are located in

```
/mnt/ppo-data-services/post-processing/annotation/<date>/<camera>/
```

- Check extracts to ensure:
    - the FO is not cut off,
    - the FO is not excessively tilted resulting in the **surface area appearing smaller** than the size it is meant to be

- there is only one FO in the annotation box
- NOTE: If a set of extracts needs to be redone, extracts no longer need to be deleted
  - Rerunning `python ~/ppo/tools/visual/review_tool_aws.py <post_processing.txt> --make extracts` and editing the affected scan will delete and replace the bad extract

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Uploading to Server

(includes all h.5 files, annotations & extracts / ROIs to S3 Bucket)

The `/mnt/ppo-experimental-data` bucket where the raw data is stored is read only, so the corrected data/ extractions must be uploaded to a separate S3 bucket `/mnt/ppo-post-proc`

- Run `python ~/ppo/tools/data_utils/aws_copy.py` or use alias `awscp` on the relevant .txt file.
  - The output directory information is encoded in the .txt file.
- Once completed, confirm the correct number of files have been moved to the server using `ls | wc -l` .
  - You will need to be in the directory where the corrected files are.
    - Ex. `/mnt/ppo-post-proc/FMK_IF/fo/25005/dm24291001_vis/2025-07-07/corrected`
  - **Make sure to check both vis & spec files have been all uploaded**
- Use extra caution that any bad scans are not moved to the server as wel

Files can then be removed from local computers' persistent volumes to make space

- Copying files with `aws_copy` will also add the metadata for those files, and add them to the appropriate list file